MILITARY SPECIFICATION

POSITION INDICATOR MARK 6 MOD 2

(FOR FIRE CONTROL SYSTEM MARK 101)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of Position Indicator Mark 6 Mod 2, Bureau of Naval Weapons ID 161944, which is used to display continuous information concerning own ship and target relative positions for the solution of submarine fire control problems.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-17555	Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-W-21927	Weapons; Handling and Preparation for Delivery of; General Requirements for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-129 Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 161944 Position Indicator Mark 6 Mod 2

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVORD OD 10212 Position Indicator Mark 6 Mod 2, Factory Acceptance Test

OP 400 General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, and drawings requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION

Z 24.3 American Standards for Sound Level Meters

Z 24.7 American Standard Test Code for Noise Measurement

Z 24.10 American Standards Association - Sound Levels

(Application for copies should be addressed to the American Society of Engineers, 29 W. 39th St., New York, N. Y.)

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

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CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION, INC. PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th St, N.W, Washington 6, D.C.)

3. REQUIREMENTS

3.1 Description. - The Position Indicator Mark 6 Mod 2 is comprised of two main units; an indicator unit which is installed in the conning tower of a submarine; and an amplifier unit which is remotely located from the indicator unit. Their arrangement permits the display of the angular relationship between own ship and target. Counters display range, torpedo run, and target speed. Dials present a visual display of true and relative target bearings, target course, own ship course, gyro angle, impact angle, and target angle. The indicator unit also includes a Forward-Aft request switch, forward and aft indicator lights, and a control for intensity of illumination.

3.2 General Requirements

3.2.1 Materials. - Unless otherwise specified in the contract or order, all materials and purchased parts entering into the final assembly of the position indicator shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUWEPS LD 161944.

3.2.2 Finishes, coatings, construction and dimensions. - Finishes, coatings, construction and dimensions for the position indicator shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD 161944.

3.2.3 Moisture, fungus and corrosion. The position indicator shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in BUWEPS LD 161944.

3.2.4 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawing listed in BUWEPS LD 161944.

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3.2.5 Interchangeability. - Individual parts of assemblies, fabricated in agreement with their applicable drawings and specifications shall be interchangeable with mating parts of the unit and other like units. If deviations greater or smaller than the tolerance shown on the drawings or specifications are required for interchangeability, or if hand fitting beyond that shown on the applicable drawing is required, then changes in dimensions are necessary and revision and correction of the applicable drawings are required. See 3.2.8.

3.2.6 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class of fit for threads shall be as specified on the applicable drawings, and the NO-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.7 Requests for deviations. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.8 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance requirements and product characteristics. - The position indicator shall meet the following performance requirements and product characteristics:

3.3.1 Dielectric and insulation resistance. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Dielectric and insulation resistance test).

3.3.2 Dial illumination. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Dial illumination).

3.3.3 Range. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Range counter).

3.3.4 Torpedo run. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Torpedo Run Counter).

3.3.5 Target speed. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Target Speed Counter).

3.3.6 Target angle. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Target Angle Dial).

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3.3.7 Target solution. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Target Solution) (A) and (B).

3.3.8 Own ship course. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Own Ship Course Dial).

3.3.9 Relative bearing. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Relative Bearing Dial).

3.3.10 Gyro angle. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Gyro Angle).

3.3.11 True bearing. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (True Bearing).

3.3.12 Impact angle. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Impact Angle).

3.3.13 Target course. - The position indicator shall be capable of meeting the requirements specified in NAVORD OD 10212 (Target Course).

3.3.14 Signal lights. - The signal lights of the position indicator shall meet the requirements specified in NAVORD OD 10212 (Signal Light).

3.3.15 Request switch. - The request switch of the position indicator shall meet the requirements specified in NAVORD OD 10212 (Request Switch).

3.3.16 Noise. - The position indicator shall not generate noise such that the arithmetical average of the sound pressure levels in the four octave bands 300-600, 600-1200, 1200-2400, and 2400-4800 cycles per second exceeds 65 decibels.

3.4 Environmental requirements

3.4.1 Inclination. - Unless otherwise specified in the contract or order, the position indicator shall be capable of withstanding the inclination tests specified in 4.5.1.

3.4.2 Vibration. - Unless otherwise specified in the contract or order, the position indicator shall be capable of withstanding vibration tests as follows:

3.4.2.1 Vibration-transportation, storage, and handling. - The position indicator, packed and packaged as defined in 5, shall be capable of withstanding tests as specified in 4.5.2.1 and thereafter shall show no damage and shall meet all the performance and product requirements of this specification.

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3.4.2.2 Vibration-operating condition. - The position indicator, self-mounted as specified in 4.5.2.2.1, shall be capable of withstanding vibration requirements as specified in 4.5.2.2.2 and thereafter shall show no damage and meet all performance requirements and product characteristics of this specification.

3.4.3 Shock. - Unless otherwise specified in the contract or order, the position indicator shall be capable of withstanding shock tests as follows:

3.4.3.1 Shock-transportation, storage, and handling. - The position indicator, packed and packaged as defined in 5 shall be capable of withstanding tests as specified in 4.5.3.1 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.3.2 Shock-operating condition. - The position indicator, mounted as specified in 4.5.3.2.1, shall be capable of withstanding shock tests as specified in 4.5.3.2.2 and thereafter shall show no damage and meet all performance requirements and product characteristics of this specification.

3.4.4 Temperature and humidity. - The position indicator shall be capable of withstanding the temperature and humidity conditions specified in 4.5.4 and 4.5.5 and thereafter shall show no corrosion or damage and meet the performance requirements and product characteristics of this specification.

3.4.5 Tropical environment. - The position indicator shall be capable of withstanding exposure to the tropical environment specified in 4.5.6 and thereafter shall show no fungus growth, corrosion, or other damage and meet the performance requirements and product characteristics of this specification.

3.4.6 Marine atmosphere environment. - The position indicator shall be capable of withstanding marine atmosphere environmental conditions specified in 4.5.7 and thereafter shall show no evidence of corrosion or damage and meet the performance requirements and product characteristics of this specification.

3.5 Clean-up. - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. - The position indicator, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting,

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riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy final inspection gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy final inspection gages.

4.1.2 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three position indicators shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Periodic production samples. - Unless otherwise specified for each contract or order, from each succeeding lot or grand lot of three hundred or less, a random sample of three position indicators will be selected by the Government inspector for submission to a Bureau of Naval Weapons designated activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The position indicator shall be considered satisfactory if the position indicator meets all specifications and pertinent drawing requirements, and withstands the tests as specified in 4.4 and 4.5 inclusive.

4.2 Classification of tests. - The inspection and testing of the position indicator shall be classified as follows:

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4.2.1 Acceptance tests. - These tests are to be accomplished on each position indicator being submitted for acceptance under contract. Acceptance tests shall either be performed by the manufacturer and witnessed by the Government inspector, or shall be conducted by the Government inspector. These tests are defined in 4.4.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.3 which are representative of the position indicator after the award of the contract to determine that the lot and production meet the requirements of this specification. These are detailed in 4.5. Failure of any sample position indicator in any test or requirement will result in the rejection of the lot or cessation of production, as determined by the procuring activity.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.3.1 Power supplies. - Tabulated below are the several sources of electrical power required:

Voltage	Frequency	Amperes (Approximate)	Power Factor (Approximate)	Volt Amperes
24	DC	--	--	--
115	60	--	--	--
24	400- 6 Phase	0.75 (Max)	0.77	18
115	400- 3 Phase	2.8	0.67	322

4.3.2 Toggle switch with two positions labelled FWD and AFT to apply 115 volts 60 cycles across terminals D4 - D3 and D3 - D5 respectively.

4.3.3 Dummy transmitters

4.3.3.1 U input. - A dummy director with a 1 and 36 speed, 24 volt, 400 cycle synchro transmitter; and 1 and 36 speed, 115 volt, 400 cycle synchro-transmitters arranged to be motor driven at constant velocities of 5 to 25 degrees per second (or thru an integrator to provide a sine wave output). A switch shall be provided so that input can be switched from one set of transmitters to the other.

4.3.3.2 cR servo. - A dummy director with one and 36 speed 115 volt, 60 cycles transmitters arranged to be motor driven at constant velocities of five to 25 degrees per second (or thru an integrator to provide a sine wave output). A 115 volt, 60 cycle reference for cR servo in the indicator unit shall be provided.

4.3.3.3 cBr servo. - A 115 volt, 60 cycle synchro transmitter. A 115 volt, 60 cycle synchro reference for cBr servo in the indicator unit shall be provided.

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4.3.3.4 Single speed units, except cBr. - 24 volt, 400 cycle (or 115 volt, 400 cycle) synchro transmitters. For "own ship course" function, the reference field of the servo motor shall be supplied with 115 volt, 400 cycle, in the indicator unit.

4.3.4 Vacuum tube voltmeter

4.3.5 Stop watch, capable of measuring 15 minutes of time with an accuracy of 0.1 second.

4.3.6 Insulation resistance meter, such as a megger, capable of measuring insulation resistance of at least three megohms at a potential of 500 volts DC.

4.3.7 Voltmeter, 1000 ohms-per-volt, capable of measuring 0.3 volts AC.

4.3.8 Phasing meter, for measuring phase angle.

4.3.9 Oscilloscope

4.4 Acceptance tests

4.4.1 The contractor or Government inspector shall conduct acceptance tests as specified in 4.2.1 to assure that the position indicator is in compliance with the requirements of 3.2, 3.3, 3.5 and 3.6.

4.4.2 Test conditions. - Unless otherwise specified, the position indicator shall be subjected to acceptance tests under the following conditions.

4.4.2.1 Temperature. - Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35.0 degrees C).

4.4.2.2 Altitude. - Normal ground.

4.4.2.3 Vibration. - None.

4.4.2.4 Humidity. - Room ambient to 95 percent relative (maximum).

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure

4.4.4.1 Dielectric and insulation resistance. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Dielectric and Insulation Resistance Test).

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4.4.4.2 Dial illumination. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Dial Illumination).

4.4.4.3 Range. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Range Counter).

4.4.4.4 Torpedo run. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Torpedo Run Counter).

4.4.4.5 Target speed. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Target Speed Counter).

4.4.4.6 Target angle. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Target Angle Dial).

4.4.4.7 Target solution. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Target Solution) (A) and (B).

4.4.4.8 Own ship course. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Own Ship Course Dial).

4.4.4.9 Relative bearing. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Relative Bearing Dial).

4.4.4.10 Gyro angle. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Gyro Angle).

4.4.4.11 True bearing. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (True Bearing).

4.4.4.12 Impact angle. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Impact Angle).

4.4.4.13 Target course. - The position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Target Course).

4.4.4.14 Signal lights. - The signal lights of the position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Signal Light).

4.4.4.15 Request switch. - The request switch of the position indicator shall be subjected to the tests specified in NAVORD OD 10212 (Request Switch).

4.5 Pre-production and periodic production tests

4.5.1 Inclination

4.5.1.1 Mounting. - The position indicator shall be mounted on the inclination testing machine in its normal operating position.

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4.5.1.2 The position indicator shall be inclined at the rate of five to seven cycles per minute in one plane to angles of 45 degrees on either side of the vertical for a period sufficiently long to determine its electrical characteristics under such motion or for a minimum of 30 minutes.

4.5.1.3 Reorient the position indicator 90 degrees and repeat the procedure specified in 4.5.1.2.

4.5.1.4 At the conclusion of the test procedures specified in 4.5.1.2 and 4.5.1.3, the cyclic motion may be stopped and the inclination adjusted to any angle or angles within 45 degrees of the vertical for sufficient time to investigate the electrical characteristics of the equipment. At the conclusion of the tests, the position indicator shall meet the performance and other requirements specified in this specification.

4.5.2 Vibration

4.5.2.1 Vibration. - Transportation, storage, and handling. - The position indicator, packaged and packed as specified in Section 5, shall be vibration tested as specified in Specification MIL-W-21927. At the conclusion of the test, the position indicator shall meet the requirements of 3.4.2.1.

4.5.2.2 Vibration, operating condition.

4.5.2.2.1 Mounting. - The position indicator shall be mounted on the vibration testing machine to simulate as realistically as possible the service installation, including vibration isolator or mounting devices. The position indicator shall be secured to the vibration testing machine in such a manner that the direction of vibration will be along each of the principle axes separately. The position indicator may be reoriented on the horizontal vibrating machine between each test, if necessary, to obtain such motion. At no time shall the equipment be mounted in any other way than its normal position.

4.5.2.2.2 Test procedures. - The position indicator shall be energized to perform its normal function in the operating condition and subjected to the following tests. If it is found necessary to repair any damage severe enough to constitute a failure of the equipment the entire test shall be repeated, unless otherwise directed by the Bureau of Naval Weapons or agency concerned.

4.5.2.2.2.1 Exploratory vibration test. - To determine the presence of resonances in the position indicator, it shall be vibrated at frequencies from five cycles per second to 33 cycles per second, at a table vibratory amplitude of 0.010 plus or minus 0.002 inch. The change in frequency shall be made in discrete frequency intervals of one cycle per second and maintained at each interval for 15 seconds. The frequencies at which resonances occur shall be noted.

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4.5.2.2.2.2. Variable frequency test. - The position indicator shall be vibrated from five cycles per second to 23 cycles per second, in discrete frequency intervals of one cycle per second. At each interval frequency, the vibration shall be maintained for three minutes. The change from one frequency to the next shall be made slowly and uniformly. Tests shall be made at a table amplitude between a minimum of 0.024 inch and a maximum of 0.036 inch.

4.5.2.2.2.3 Endurance test. - The position indicator shall be vibrated for a total period of at least two hours, at the resonant frequencies selected by the test engineer. If no resonance was observed, this test shall be performed at 23 cycles per second. At the conclusion of the tests specified by 4.5.2.2.2, the position indicator shall meet the performance and other requirements specified in this specification.

4.5.3 Shock

4.5.3.1 Shock - Transportation, storage and handling. - The position indicator, packaged and packed as specified in Section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927. At the conclusion of the tests, the position indicator shall meet the requirements of 3.4.3.1.

4.5.3.2 Shock-operating condition

4.5.3.2.1 Mounting. - The position indicator shall be attached to the anvil table by means of flexible steel mounting brackets. The position indicator shall be so mounted as to simulate as realistically as possible the service installation including shock mounts.

4.5.3.2.2 Test procedure. - The position indicator shall be energized to perform its normal function in the operating condition and subjected to the following test. If it is found necessary to repair any damage severe enough to constitute a failure of the equipment, the entire test shall be repeated, unless otherwise directed by the Bureau of Naval Weapons or agency concerned.

4.5.3.2.2.1 Shock test. - The shock test shall consist of six blows applied in three groups of two blows each. The height of the hammer drop shall be nine inches for the first group and 21 inches for the second and third groups of blows. The up travel of the anvil table shall be three inches for groups one and two and one and one half inches for the third group. At the conclusion of this test, the position indicator shall meet the performance and other requirements specified in this specification.

4.5.4 Temperature test

4.5.4.1 Procedure I. - The position indicator shall first be placed within a test chamber wherein a temperature of 85 degrees C (185 degreesF) is maintained. The equipment shall be maintained at this temperature for a period of four hours, at the conclusion of which, and within five minutes,

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the equipment shall be transferred to a chamber having an internal temperature of minus 40 degrees C (minus 40 degrees F). The equipment shall be subjected to this temperature for a period of four hours. The number of complete cycles shall be three. At the conclusion of the third cycle the equipment shall be removed from the test chamber, and operated within a period of one hour. The position indicator shall then meet the performance and other requirements specified in this specification. Deterioration or corrosion of any internal or external components which could in any manner prevent the equipment from meeting operational requirements during service life shall provide reason to consider the equipment as having failed to meet the test.

4.5.4.2 Procedure II. - The position indicator shall be placed within the chamber and maintained for a period of at least one hour or until the equipment performance stabilizes at a temperature of 25 degrees plus or minus 15 degrees C (77 degrees plus or minus 27 degrees F). The chamber temperature shall then be reduced to minus 55 degrees C (minus 67 degrees F) and maintained at this condition for at least one hour. The internal temperature of the chamber shall then be increased to 71 degrees C (160 degrees F) and maintained at this temperature for at least one hour. The internal temperature shall then be returned to 25 degrees plus or minus 15 degrees C (77 degrees plus or minus 27 degrees F). The position indicator shall then meet the performance and other requirements specified in this specification. Deterioration or corrosion of any internal or external components which could in any manner prevent the equipment from meeting operational requirements during service life shall provide reason to consider the equipment as having failed to meet the test. Tests shall be made at any other ambient temperature between the limits specified, at the option of the qualifying agency or as stated in the detail specification.

4.5.5 Humidity tests

4.5.5.1 Procedure 1. - The position indicator shall be placed in a test chamber and set up to simulate installed conditions. Distilled or demineralized water having a pH value of between 6.5 and 7.5 at 25 degrees C (77 degrees F) shall be used to obtain the desired humidity. The temperatures in the chamber shall be raised from 29 degrees plus or minus nine degrees C (84.2 degrees plus or minus 16.2 degrees F) to 71 degrees C (160 degrees F) during a two hour period. The temperature of 71 degrees C (160 degrees F) and a relative humidity of 95 percent shall be maintained during the next six hour period. During the following 16-hour period, the temperature in the chamber must drop at a uniform rate to 29 degrees plus or minus nine degrees C (84.2 degrees plus or minus 16.2 degrees F) which constitutes one cycle. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 360 hours (15 cycles) or as specified by the detail specification. At the conclusion of the 360 hour period, the position indicator shall be inspected for evidence of deterioration or corrosion and shall then meet the performance and other requirements specified in this specification.

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4.5.5.2 Procedure II. - The equipment shall be placed in a chamber having a temperature of 49 degrees C (120 degrees F) and humidity of 95 percent. The test conditions shall be maintained for 360 hours. At the conclusion of this period, the position indicator shall be operated and shall meet the performance and other requirements specified in this specification.

4.5.6 Tropical environment. - The position indicator shall be subjected to following tropical environmental test with all access covers removed. Drying or wiping prior to performance checking shall not be done.

4.5.6.1 Test procedure. - The position indicator shall be placed in a mold chamber maintaining an internal temperature of 30 degrees C (86 degrees F) and a relative humidity of 95 percent and sprayed with the suspension of mixed spores. These conditions shall be maintained for 28 days. At the conclusion of the test the position indicator shall be allowed to return to room temperature before performance checking. A list of five groups of fungi are listed below. One type of fungi from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at 25 degrees C) (77 degrees F) shall be prepared by sterilization in convenient containers (approximately 100 milliliters each). Approximately 10 milliliters of the solution shall then be introduced directly into a stock culture of one fungus and shaken vigorously in order that a well sporulated suspension will result without disturbing the agar. This process shall be repeated for each type of fungus. The separate spore suspensions from the types of fungi shall then be mixed together in an atomizer to provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no condition will the inoculum be kept for more than an eight hour period.

GROUP I	Chaetomium globosum - USDA 1042.4 or Myrothecium verrucaria - USDA 1334.2.
GROUP II	Rhizopus nigricans - S.N. 32 or Aspergillus niger - USDA TC 215-4247.
GROUP III	Aspergillus flavus - WADC No. 26 or Aspergillus terreus - PQMD 82J.
GROUP IV	Penicillium luteum - USDA 1336.1 Penicillium sp. - USDA 1336.2 or Penicillium citrinum - ATCC 9849
GROUP V	Menoniella echinata - WADC No. 37 or Fusarium moniliforme - USDA 1004.1

4.5.7 Marine atmosphere environment. - When specified by the procuring agency, the position indicator shall be subjected to the following sea atmosphere conditions.

4.5.7.1 The position indicator assembled, sealed, and ready for service use, shall be mounted in the exposure chamber maintained at 35 degrees C (95 degrees F). A salt solution (see 4.5.7.1.2) spray creating a finely

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divided, wet dense fog shall be introduced into the chamber. The conditions maintained in all parts of the exposure zone shall be such that a suitable receptacle placed at any point in the exposure zone will collect from 0.5 to three ml of solution per hour for each 80 square cm of horizontal collecting area (10 cm diameter) based on an average of at least 16 hours. The length of the marine atmosphere test shall be not less than 50 hours.

4.5.7.1.2 Salt solution. - The salt used shall be sodium chloride containing on the dry basis not more than 0.1 percent of sodium iodide and not more than 0.2 percent of total impurities. The solution shall be prepared by dissolving 20 plus or minus two parts by weight of salt in 80 parts by weight of distilled or other water containing not more than 200 parts per million of total solids. The solution shall be adjusted to and maintained at a specific gravity of from 1.126 to 1.157 and at a pH of between 6.5 to 7.2 when measured at a temperature between 33 degrees and 36 degrees C (92 degrees and 97 degrees F). Only C.P. hydrochloric acid or C.P. sodium hydroxide shall be used to adjust the pH.

4.5.7.2 At the conclusion of the marine atmosphere test, the position indicator shall meet the performance and other requirements specified in this specification. Salt deposits resulting from the test may be removed prior to operation.

4.5.8 Noise. - The generated noise level shall be measured by the test procedures described in the Current American Test Code for Noise Measurement, Z24.7 - 1950, except that measurements shall be made with both a sound level meter conforming to American Standard Z24.3 - 1944, and an octave band filter set conforming to American Standard Z24.10. At each microphone station, an octave band analysis of the noise shall be made.

4.6 Waiver of tests. - Tests specified in 4.5.2.1 and 4.5.3.1 may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Failure of the position indicator to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, drying, and preservative application. - Not applicable.

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5.1.1.2 Unit packaging. - The indicator unit and amplifier unit of the position indicator shall be individually packaged in accordance with Method II-B of Specification MIL-P-116.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the position indicator will meet the requirements of Section 3 after being tested in accordance with Section 4.

5.1.2 Level B. - Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit packaging. - The indicator unit and amplifier unit of the position indicator shall be individually packaged in accordance with Method III of Specification MIL-P-116.

5.1.3.3 Intermediate package. - Not applicable.

5.2 Packing

5.2.1 Level A. - In addition to the following, Level A packing shall be in accordance with Specification MIL-P-17555.

5.2.1.1 Exterior containers. - One position indicator, packaged as described in 5.1.1.2, shall be packed in an overseas type wood box in accordance with Specification PPP-B-601 or PPP-B-621.

5.2.2 Level B. - In addition to the following, Level B packing shall be in accordance with Specification MIL-P-17555.

5.2.2.1 Exterior containers. - One position indicator, packaged as described in 5.1.1.2, shall be packed in a domestic type box in accordance with Specification PPP-B-601 or PPP-B-621.

5.2.3. Level C. - Position indicators, packaged as described in 5.1.3.2, shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or Motor Freight Classification Rules, whichever is applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal marking. - In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129, including serial numbers.

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5.3.2 Special marking. - None, unless otherwise specified.

6. NOTES

6.1 Intended use. - Position Indicator Mark 6 Mod 2 is intended for use in Fire Control System Mark 101.

6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and other documents. Section 5 should be adhered to for selection of applicable levels of preservation, packaging, and packing. Selection of Level C shall not be made unless immediate use is definitely known.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and Options in Specification MIL-P-116 and to paragraph 4.6 and paragraph 5.3.1.

6.3 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less, or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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